Product Backlog

Team Name: S.E.N.D.
Project Title: Marketplace Product Price Comparator App
Course: CIS 553 - 002, Fall 2022
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❖ Refine your user stories considering the instructor's feedback. Break down previously identified large user stories. (2 pts)

ANS: There were originally three user stories as in the Table 1 below. All those user stories were identified as large and complex stories and we split those into smaller ones. The broken-down stories into smaller Tasks which are listed in Table 2 below

Table 1						
Epic	**Issue Type**	**Issue key**	**Issue id**	**Summary**	**Description**	**Story Point**

SEND-16	Story	SEND-17	10016	Develop a User Interface for the App for searching product by keyword	As a user of the “Marketplace Product Price Comparator App”, I want to be able to search the product using a keyword through a Graphical User Interface (GUI), so that I can get the price and other attributes of the product from different marketplace sites. <u>Pre-conditions:</u> The search box in the GUI is available and empty to be able to enter search keyword <u>Post-conditions:</u> Customer receives the compared list of attributes of the desired product	8
SEND-16	Story	SEND-18	10017	Develop web scraper module to search and display product attributes on GUI	As a user of the “Marketplace Product Price Comparator App”, I want to be able to see the search result list (product attributes) from different marketplace sites on a Graphical User Interface (GUI) , so that I can compare them to find the best match (e.g., best price, best rating, etc.). <u>Pre-conditions:</u> The search result fields (in a table form) in the GUI is available The web scraping of the product attributes from all the marketplace sites are complete <u>Post-conditions:</u> The list of product attributes along with the names of the marketplace sites (un-sorted list) is displayed on the GUI	13

SEND-16	Story	SEND-19	10018	Develop a User Interface for the App to sort the search results	As a user of the “Marketplace Product Price Comparator App”, I want to be able to sort the search results for attributes of a product from various marketplace sites using a option through a Graphical User Interface (GUI), so that I can get list sorted by desired attribute (e.g., price, ratings etc.). <u>Pre-conditions:</u> The drop down menu in the GUI is available to select the attribute that user would like to sort the list by <u>Post-conditions:</u> Customer receives the sorted compared list of attributes of the desired product	5

Table 2						
Issue Type	**Issue key**	**Parent Issue**	**Issue id**	**Summary**	**Description**	**Story Point**
Epic	SEND-16	Not Applicable	10015	User stories based on system use cases	Create user stories based on the use cases defined in the systems requirements located on google drive here	Not Applicable

Story	SEND-17	SEND-16	10016	Develop a User Interface for the App for searching product by keyword	As a user of the Marketplace Product Price Comparator App, I want to be able to search the product using a keyword through a Graphical User Interface (GUI), so that I can get the price and other attributes of the product from different marketplace sites . PRE-CONDITIONS: # The set up for Python code development (IDE, interpreter etc.) is ready # The required Python libraries have been identified and installed POST-CONDITIONS: The UI provides interface to search the product, to select product attribute for sorting the results and display the results (sorted and unsorted)	8
Task	SEND-21	SEND-17	10020	Create a User Interface framework in Python	Create a user interface framework in Python PRE-CONDITIONS: # The set up for Python code development (IDE, interpreter etc.) is ready # The required Python libraries have been identified and installed POST-CONDITIONS: The UI framework is working and tested with some simple user interfaces (e.g., a box, button etc.)	3

Task	SEND-22	SEND-17	10021	Add user selection options in the user interface	Add following options in the user interface framework: # Text box to enter the keyword for searching the product # Select all the marketplace sites by clicking on the checkboxes against their names # Click button to start search PRE-CONDITIONS: The framework for the UI is ready and working without any error. POST-CONDITIONS: The options listed above shall be available and working without any problem.	3
Story	SEND-18	SEND-16	10017	Develop web scraper module to search and display product attributes on GUI	As a user of the Marketplace Product Price Comparator App, I want to be able to see the search result list (product attributes) from different marketplace sites on a Graphical User Interface (GUI) , so that I can compare them to find the best match (e.g., best price, best rating, etc.). PRE-CONDITIONS: * The libraries to use for webscraping is identified and installed * The investigation on what the product attributes to scrape from all the marketplace sites is complete POST-CONDITIONS:	13

					The list of product attributes along with the names of the marketplace sites (un-sorted list) is displayed on the GUI	
Task	SEND-23	SEND-18	10022	Create a framework for webscraping a product from a marketplace site	Create a framework in Python using Beautifulsoup and/or Selenium libraries to scrape a marketplace site for searched product's attributes. PRE-CONDITIONS: * The libraries to use for webscraping is identified and installed * The investigation on what the product attributes to scrape from all the marketplace sites is complete POST-CONDITIONS: The framework for the webscraping is available with following features: # It can successfully load the marketplace site # It can search a product	5

Task	SEND-24	SEND-18	10023	Create flexibility for adding marketplace sites in webscraping code	Using framework, add flexibility (e.g., class, methods etc.) that can be used to add more marketplace sites PRE-CONDITIONS: The framework for the webscraping is available with following features: * It can successfully load the marketplace site * It can search a product POST-CONDITIONS: The webscraping framework can add as many as needed marketplace sites search a product across all the sites and get desired result	3
Task	SEND-25	SEND-18	10024	Add functionality to webscraping framework to read product attributes	Add code to read out the product attributes to previously created webscraping framework. PRE-CONDITIONS: The framework for the webscraping is available with following features: * It can successfully load the marketplace site * It can search a product * It search products in several different market place sites POST-CONDITIONS: The webscraping code shall return all desired product attributes	3

Task	SEND-26	SEND-18	10025	Results table in a data structure form	The product search result (see following) shall be stored in a data structure that can be used by the UI to display the results: * Product name * Marketplace site name * Product Attribute 1 * Product Attribute 2…. * Product Attribute n PRE-CONDITIONS: The webscraping code shall return all desired product attributes POST-CONDITIONS: The product details along with the product attributes shall be returned in a table (data structure) form	5
Story	SEND-19	SEND-16	10018	Develop a User Interface for the App to sort the search results	As a user of the Marketplace Product Price Comparator App, I want to be able to sort the search results for attributes of a product from various marketplace sites using a option through a Graphical User Interface (GUI), so that I can get list sorted by desired attribute (e.g., price, ratings etc.). PRE-CONDITIONS: The drop down menu in the GUI is available to select the attribute that user would like to sort the list by	5

					POST-CONDITIONS: Customer receives the sorted compared list of attributes of the desired product	
Task	SEND-27	SEND-19	10026	Create an option in UI to sort results	Create an option in UI that can allow the user to choose the attribute that is used to sort the results PRE-CONDITIONS: The drop down (or similar) menu in the GUI is available to select the attribute that user would like to sort the list by POST-CONDITIONS: The user selected product attribute for sorting shall be captured in a UI backend variable with correct value	3
Task	SEND-28	SEND-19	10027	Sort the result in the background	Sort the result table in the background and store in the same or a different data structure PRE-CONDITIONS: The user selection for the sorting by product attribute is available POST-CONDITIONS: Based on the user selected product attribute for sorting, the result table shall be sorted and the sorted table is stored in the same or a new data structure	5

Task	SEND-29	SEND-19	10028	Display sorted results on the UI	Pull the sorted results from the data structure and display on the UI PRE-CONDITIONS: Based on the user selected product attribute for sorting, the result table shall be sorted and the sorted table is stored in the same or a new data structure POST-CONDITIONS: The sorted table from the data structure shall be displayed on the UI	3

- Estimate the size of your user stories. Use Fibonacci numbers to represent a relative size of each user story. Note the cumulative size of all user stories in your product backlog. (2 pts)

ANS: See the column "Story Points" in Table 2 above where we allocated story points that use Fibonacci numbers to represent the size of the stories.

- Provide an updated numbered list of all user stories; indicate pre- and post-conditions.

ANS: Table 2 above provides an updated numbered list of all user stories (Epic, Parent User Stories and Tasks). The pre- and post-conditions are in the Description field.

- Considering the pre- and post-conditions, identify a subset of user stories to be implemented during the first iteration (there will be a total of three iterations). Be sure that the cumulative size of the selected user stories is about 1/3 of the size of the full backlog. Describe the functionality that your partially implemented system will have at the end of this iteration. (3 pts)

ANS: Total of 3 stories (in Table 3) have been identified in the first iteration of product development. The total size of backlog (i.e., all the tasks to complete) is 33 and total size of the selected stories is 11 which is matching with the criteria of 1/3 of total backlog.

At the end of this first iteration, team is expecting to have following partial functionalities:

1) **The User Interface framework is ready and tested for a simple input**
2) **The framework for the webscraping is available which can successfully load the marketplace site and search a product**
3) **Flexibility in webscraping code to add as many as needed marketplace sites**

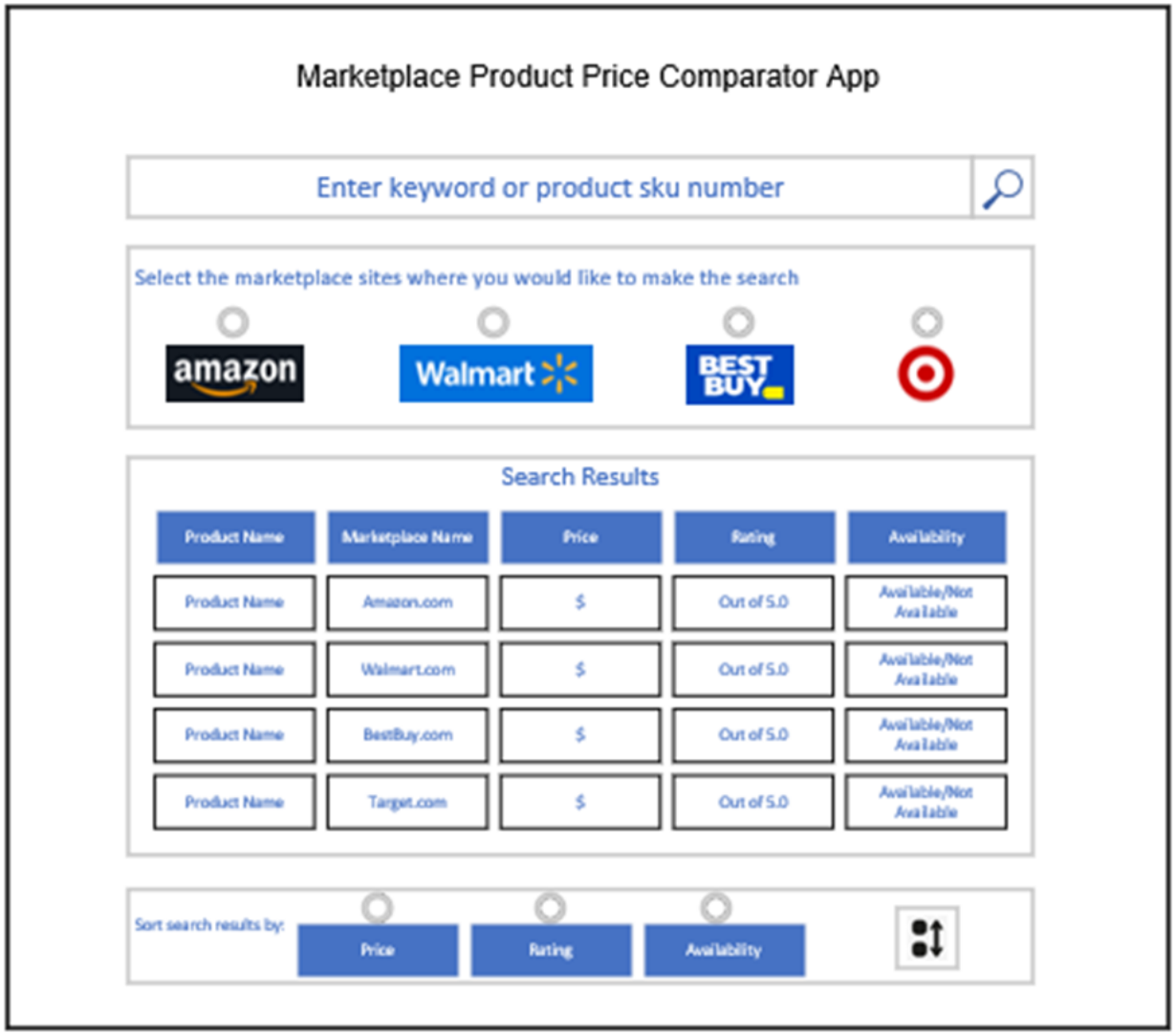
NOTE: For SEND team, the Sprint 1 was the research and investigation work. The actual product development starts with Sprint 2 which is the first iteration.

Table 3 (Sprint 2: October 26 to Nov 9, 2022)						
Issue Type	**Issue key**	**Parent Issue**	**Issue id**	**Summary**	**Description**	**Story point estimate**
Task	SEND-21	SEND-17	10020	Create a User Interface framework in Python	Create a user interface framework in Python PRE-CONDITIONS: # The set up for Python code development (IDE, interpreter etc.) is ready # The required Python libraries have been identified and installed POST-CONDITIONS: The UI framework is working and tested with some simple user interfaces (e.g., a box, button etc.)	3

Task	SEND-23	SEND-18	10022	Create a framework for webscraping a product from a marketplace site	Create a framework in Python using Beautifulsoup and/or Selenium libraries to scape a marketplace site for searched product's attributes. PRE-CONDITIONS: * The libraries to use for webscraping is identified and installed * The investigation on what the product attributes to scrape from all the marketplace sites is complete POST-CONDITIONS: The framework for the webscraping is available with following features: # It can successfully load the marketplace site # It can search a product	5
Task	SEND-24	SEND-18	10023	Create flexibility for adding marketplace sites in webscraping code	Using framework, add flexibility (e.g., class, methods etc.) that can be used to add more marketplace sites PRE-CONDITIONS: The framework for the webscraping is available with following features: * It can successfully load the marketplace site * It can search a product POST-CONDITIONS: The webscraping framework can add as many as needed marketplace sites search a product across all the sites and get desired result	3

❖ Design key features of the user interface; provide sketches of your designs. (2 pts)

ANS: Following design shows the features for the user interface for the "Marketplace Product Price Comparator App":



Additionally, here is a different iteration of the UI for the "Marketplace Product Price Comparator App". Both options may be viable and components may be used from both of them

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- Checklist for this deliverable:
  - Estimated number of hours by each member:

Hours Spent on	Name of the members			
	Dheeraj Mohanbabu	Steve Anderau	Nayan Patel	Ethan Pellegrini
Week of 10/10	3	4	2	3.5
Week of 10/17	3	1	3.5	3
Week of 10/24	3	3	5.0	2
Total Hours	9	8	9.5	8.5

  - Meeting dates since the last project deliverable:

    10/13 at 8:00 PM-9:00 PM

    10/20 at 8:00 PM-9:00 PM